Assignments

2D Arrays - 1

Q1. Write a program to store 10 at every index of a 2D matrix with 5 rows and 5 columns.

Solution:

```
#include<iostream>
using namespace std;
int main(){
int matrix[5][5];
for(int i=0;i<5;i++){
for(int j=0;j<5;j++){
matrix[i][j] = 10;
}
}
for(int i=0;i<5;i++){
for(int j=0;j<5;j++){
cout << matrix[i][j] << " ";
}
cout<<endl;
}
}
```

Q2. Write a program to add two matrices and save the result in one of the given matrices.

Input 1:

1 2 3

4 5 6

7 8 9

4 5 8

0 0 8

1 2 0

Output 1:

5 7 11

4 5 14

8 10 9

Solution :

```
#include<iostream>
using namespace std;
int main(){
int n , m;
```

```
cout << "Enter the number of rows : ";
cin >> n;
cout << "Enter the number of columns : ";
cin >> m;
int a[n][m];
cout << "Enter the first matrix : "<<endl;
for(int i = 0 ; i < n ; i++){
for(int j = 0 ; j < m ; j++){
cin >> a[i][j];
}
}
int b[n][m];
cout << "Enter the second matrix : "<<endl;
for(int i = 0 ; i < n ; i++){
```

Q3. Given a matrix 'A' of dimension n x m and 2 coordinates (l1, r1) and (l2, r2). Return the sum of the rectangle from (l1,r1) to (l2, r2).

Input 1:

1 2 -3 4
0 0 -4 2
1 -1 2 3
-4 -5 -7 0
l1 = 1, r1 = 2 , l2 = 3 , r2 = 3

Output 1:

-4

Input 2:

1 2 -3 4
0 0 -4 2
1 -1 2 3
-4 -5 -7 0
l1 = 1, r1 = 0 , l2 = 0 , r2 = 3

Output 2:

2

Solution :

```
#include<iostream>
using namespace std;
int main(){
int n,m;
cout << "Enter the number of rows : ";
cin >> n;
```

```
cout << "Enter the number of columns : ";
cin >> m;
int a[n][m];
cout << "Enter the matrix element : ";
for(int i = 0 ; i < n ; i++){
for(int j = 0 ; j < m ; j++){
cin >> a[i][j];
}
}
int l1 , l2 , r1 , r2;
cout << "Enter the value of l1 coordinate : ";
cin >> l1;
```

Q4. Write a C++ program to find the largest element of a given 2D array of integers.

Input 1:

1 3 4 6
2 4 5 7
3 5 6 8
4 6 7 9

Output 1:

9

Solution :

```
#include<iostream>
using namespace std;
int main(){
int n , m;
cout << "Enter the number of rows : ";
cin >> n;
cout << "Enter the number of columns : ";
cin >> m;
int a[n][n];
cout << "Enter the matrix elements : ";
for(int i = 0 ; i < n ; i++){
for(int j = 0 ; j < m ; j++){
cin >> a[i][j];
}
}
int maximum = -1000000;
for(int i = 0 ; i < n ; i++){
```

Q5. Write a program to print the row number having the maximum sum in a given matrix.

Input 1:

1 3 5 7
3 4 7 8
1 4 12 3

Output 1:

2

Explanation : The 2nd row has the maximum sum i.e. 1+4+12+3 = 20

Solution :

```
#include<iostream>
using namespace std;
int main(){
int n , m;
cout << "Enter the number of rows : ";
cin >> n;
cout << "Enter the number of columns : ";
cin >> m;
int a[n][n];
cout << "Enter the matrix elements : ";
for(int i = 0 ; i < n ; i++){
for(int j = 0 ; j < m ; j++){
cin >> a[i][j];
}
}
int maximum = -1000000;
int rowNumber = -1;
```

Q6. Write a function which accepts a 2D array of integers and its size as arguments and displays the elements of middle row and the elements of middle column.

[Assuming the 2D Array to be a square matrix with odd dimensions i.e. 3x3, 5x5, 7x7 etc...]

Input 1:

1 2 3 4 5
3 4 5 6 7
7 6 5 4 3
8 7 6 5 4
1 2 37 8 0

Output 1:

3
5
7 6 5 4 3
6
37

Solution :

```
#include<iostream>
using namespace std;
int main(){
int n;
cout << "Enter the number of rows : ";
cin >> n;
int a[n][n];
cout << "Enter the matrix elements : ";
for(int i = 0 ; i < n ; i++){
for(int j = 0 ; j < n ; j++){
cin >> a[i][j];
}
}
cout << "The elements of the middle row and middle column are as
follows :
"<<endl;
int i,j;
for(i = 0 ; i < n ; i++){
for(int j = 0 ; j < n ; j++){
if(i == n/2 or j == n/2)cout<<a[i][j]<<" ";
```

THANK YOU !